

Muhammad Farooqi

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EDUCATION

University of Waterloo

Mechatronics and Robotics Engineering + AI Option — Grade: **91%**

Relevant Courses: OOP in C/C++, Data Structures and Algorithms, Digital Logic, Linear Circuits, Linear Algebra.

Waterloo, ON

Expected Apr-2028

TECHNICAL SKILLS

Programming Languages: Python, C++, Rust, TypeScript, JavaScript, SQL, HTML/CSS

AI & ML: PyTorch, OpenCV, NumPy, ONNX, TensorRT, YOLOv8/v11, HuggingFace, MediaPipe, nuScenes

Robotics & Simulation: ROS2, Isaac Sim, Foxglove, Rviz, Gazebo, Isaac Lab, Mujoco, SLAM, Arduino, Raspberry Pi

Developer Tools: Git, Docker, Linux, CUDA, Jupyter Notebook, Claude, Cursor, MCP, Figma, Ollama

EXPERIENCE

Navigation Software Engineering Intern

Sep 2025 – Present

Rivian

Vancouver, BC

- Owned integration test planning for the vehicle Driver Display swap feature, creating **40** test cases covering **ADAS** and map view transitions, button press sequences, and map state changes, catching **5** critical blockers before release.
- Automated **40** navigation test flows in nightly **CI**, covering reroutes, destination edits, and driver display changes.
- Pioneered an **APK** testing pipeline, triggering downstream automated smoke tests on PR's, replacing a **3-hour** XMM image build with a **30-min** APK and catching issues before **QA**, preventing regressions in **20%** of daily MRs.
- Extended the **APK** pipeline with **Claude AI** agents using **Jira** and **GitLab** context and a custom-built **ADB MCP** to run targeted tests, enabling self-service testing and sending higher-quality MRs to **QA**.

Autonomy Software Engineer

Feb 2025 – Present

WATOnomous

Waterloo, ON

- Built a **ROS2** world-mapping node converting **Intel D455** depth frames to 3D point clouds using camera intrinsics, then voxelized with **spconv** PointToVoxel, achieving **4x** speedup with stride downsampling, visualized in **Foxglove**.
- Accelerated depth estimation with **DepthAnythingV2** in **perception** stack by exporting **PyTorch** weights to **ONNX** and compiling a **TensorRT** engine, cutting inference latency **29%** and boosting inference speed from **85 to 120 FPS**.
- Built a **PyTorch** VAE on **nuScenes** data to detect camera frames reconstruction error for vehicle camera health.
- Implemented a **DLS IK** solver in **Isaac Sim** using **MuJoCo**, converging a 21-DOF hand to **0.1mm** tolerance.
- Fine-tuned **ACT/GR00T-N** policies on an SO-100 arm for letter-drawing task via **Imitation Learning** in **LeRobot**.

Robotics Software Engineer Intern

May 2025 - Aug 2025

Lincoln Electric Automation

Waterloo, ON

- Designed and deployed an end-to-end tack welding program for 6-axis **Fanuc** (KAREL) and **ABB** (RAPID) arms, removing separate welder setup and saving **5–7 seconds** per tack, **2–3 minutes** per part across **2** production cells.
- Developed motion control and weld parameter interfaces in **TypeScript** using **functional programming**, integrating with **Controller APIs** for 3 tack sizes and validating through **100+** virtual welds in **RobotStudio** and **RoboGuide**.

Backend Software Developer Intern

Sep 2024 – Dec 2024

ISARA

Waterloo, ON

- Built cryptographic risk analysis algorithms for TLS, SMB, and plaintext protocols using **Node.js** and **GraphiQL**.
- Collaborated with QA to resolve **355+** failures during regression testing, improving product health by **88.75%**.
- Automated demo deployment using a **Python** script to fetch build from an **AWS S3** bucket and developed an admin control panel with **React**, **PostgreSQL**, and **Docker**, saving **10 minutes** per deployment.

Computer Vision Intern

Jan 2024 – Apr 2024

AirMatrix

Mississauga, ON

- Annotated over **3,000** traffic images using **Roboflow**, achieving **95%** accuracy at **0.5 IoU** on vehicle detection.
- Trained a **YOLOv8** vehicle detection model in **Python** for vehicle detection, counting, and speed estimation.

PROJECTS

SLAM ROS2 Robot | *Linux, ROS2, Python, Gazebo, Rviz, Camera/Lidar, Raspberry Pi, Arduino*

- Created **URDF** files with collision and inertia values for realistic simulation in **ROS2** and **Gazebo**.
- Configured a differential drive robot in **Rviz** with **camera** and **2D LiDAR** nodes, generating point clouds from LiDAR and applying **multi-sensor** fusion with **SLAM Toolbox** for mapping and localization.
- Integrated motor control using **Raspberry Pi 4** and **Arduino** over **UART** at **57600 baud**, and enabled teleoperation with a **PS4 controller** via the **joy** topic using the **ros2 control** library.

